

Jin Lei^{PH.D.}

Professor, Tongji University
Shanghai 200092, China

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EMPLOYMENT HISTORY

- 2026.5 – 2027.10 ♦ **Visiting Scientist** (concurrent) Southern Center for Nuclear-Science Theory (SCNT), Institute of Modern Physics, Chinese Academy of Sciences, Huizhou
- 2023.2 – Now ♦ **Professor** Tongji University
- 2020.11 – 2023.2 ♦ **Research Fellow** Tongji University
- 2019.11 – 2020.10 ♦ **Post-Doctoral Research Associate** INFN, Sezione di Pisa
Supervisor: Dr. Angela Bonaccorso
- 2016.8 – 2019.8 ♦ **Post-Doctoral Research Associate** Department of Physics and Astronomy, Ohio University
Supervisor: Prof. Charlotte Elster

EDUCATION

- 2013.10 – 2016.7 ♦ **Ph.D., University of Seville, Spain** in Theoretical Nuclear Physics.
Supervisor: Prof. Antonio M. Moro
Thesis title: *Study of Inclusive Breakup Reactions Induced by Weakly Bound Nuclei*.
More details at <https://idus.us.es/xmlui/handle/11441/44344>
- 2010.9 – 2013.7 ♦ **M.Sc., University of Chinese Academic of Sciences, China** in Nuclear Physics
Supervisor: Prof. Jiansong Wang
Thesis title: *Reduction Method for Low-energy Nuclear Reaction Systems*.
- 2006.9 – 2010.7 ♦ **B.Eng., Northeastern University, China.** in Mechanical Engineering

SKILLS

- Languages ♦ Native speaker of Chinese; strong reading, writing and speaking competencies in English.
- Coding ♦ Fortran (incl. CUDA Fortran), C/C++, Python (PyTorch, JAX), Julia, CUDA, $\LaTeX$

RESEARCH PUBLICATIONS

Journal Articles

- 1 Hao Liu, **Jin Lei** and Zhongzhou Ren, ‘Coherent absorption dynamics: the dual role of off-diagonal couplings in weakly bound nuclei’, *Phys. Rev. C* **113**, 054601 (2026).
- 2 **Jin Lei**, ‘Reduced basis emulator for elastic scattering in continuum-discretized coupled-channels calculations’, *Phys. Rev. C* **113**, 044610 (2026).

- 3 **Jin Lei**, 'Direct boundary matching : a bound-state technique for nuclear scattering with lagrange-legendre functions', *Phys. Rev. C* **113**, 024614 (2026).
- 4 Hao Liu, **Jin Lei** and Zhongzhou Ren, 'Channel couplings redirect absorbed flux from peripheral loss to fusion in weakly bound nuclear reactions', *Physics Letters B*, 140479 (2026).
- 5 **Jin Lei**, 'Numerical assessment of convergence in the post-form ichimura-austern-vincent model', *Phys. Rev. C* **112**, 014609 (2025).
- 6 G. Yang, K. Wang, H. Liu, W. D. Chen, Y. Y. Yang, F. F. Duan, **Jin Lei**, D. Y. Pang, Z. H. Gao, S. Y. Jin, J. S. Wang, X. Liu, S. W. Xu, J. B. Ma, P. Ma, Z. Bai, Q. Hu and Z. Y. Sun, 'Direct measurement of $^8\text{B} + p$ and $^7\text{Be} + 2p$ cross sections for $^9\text{C} + ^{208}\text{Pb}$ at 303 mev', *Phys. Rev. C* **111**, 064602 (2025).
- 7 **Jin Lei**, 'Continuum effects and the trojan horse mechanism in halo nuclei-induced reactions: implications for heavy isotope synthesis', *Phys. Rev. C* **111**, 034610 (2025).
- 8 Hao Liu, **Jin Lei** and Zhongzhou Ren, 'Kolmogorov-arnold networks in nuclear binding energy prediction', *Phys. Rev. C* **111**, 024316 (2025).
- 9 Kyle Godbey, Martin Albertsson, Jacklyn M. Gates, Kris Hagel, Jesus Lubian, Andrzej Makowski, Patrick McGlynn, Gregory Potel, Jennifer L. Pore, Jeannie Rangel Borges, Kazuyuki Sekizawa, Filomena Nunes, Kaitlin Cook, Kouichi Hagino, Michal Kowal, **Jin Lei**, Matthew R. Mumpower, Witold Nazarewicz and Sait Umar, 'Paths to superheavy nuclei', *Journal of Physics G: Nuclear and Particle Physics*, Accepted Manuscript (2025).
- 10 **Jin Lei**, 'Research progress on the reaction mechanism of halo nuclei based on quantum few-body theory', *XXXX* **70**, 3270–3277 (2025).
- 11 Junzhe Liu, **Jin Lei** and Zhongzhou Ren, 'Coloss: complex-scaled optical and coulomb scattering solver', *Computer Physics Communications* **311**, 109568 (2025).
- 12 K. Wang, Y. Y. Yang, **Jin Lei**, A. M. Moro, V. Guimarães, J. G. Li, F. F. Duan, Z. Y. Sun, G. Yang, D. Y. Pang, S. W. Xu, J. B. Ma, P. Ma, Z. Bai, Q. Liu, J. L. Lou, H. J. Ong, B. F. Lv, S. Guo, M. Kumar Raju, X. H. Wang, R. H. Li, X. X. Xu, Z. Z. Ren, Y. H. Zhang, X. H. Zhou, Z. G. Hu and H. S. Xu, 'Elastic scattering and breakup reactions of the mirror nuclei ^{12}B and ^{12}N on ^{208}Pb using ab initio structure inputs', *Phys. Rev. C* **109**, 014624 (2024).
- 13 Junzhe Liu, **Jin Lei** and Zhongzhou Ren, 'A complex scaling method for efficient and accurate scattering emulation in nuclear reactions', *Physics Letters B* **858**, 139070 (2024).
- 14 G. Villanueva, A.M. Moro, J. Casal and **Jin Lei**, 'Neutron-transfer induced breakup of the borromean nucleus 9be ', *Physics Letters B* **855**, 138766 (2024).
- 15 **Jin Lei** and Antonio M. Moro, 'Advancing the ichimura-austern-vincent model with continuum-discretized coupled-channels wave functions for realistic descriptions of two-body projectile breakup', *Phys. Rev. C* **108**, 034612 (2023).
- 16 Junzhe Liu, **Jin Lei** and Zhongzhou Ren, 'Testing the validity of the surface approximation for reactions induced by weakly bound nuclei with a fully quantum-mechanical model', *Phys. Rev. C* **108**, 024606 (2023).
- 17 Yazhou Lu, **Jin Lei** and Zhongzhou Ren, 'Systematic single-folding optical potential for ^6Li and ^7Li based on $k\text{doz}$ potentials', *Phys. Rev. C* **108**, 024612 (2023).
- 18 Hao Liu, Shinsuke Nakayama, **Jin Lei** and Zhongzhou Ren, 'Comparison of ichimura-austern-vincent and glauber models for the deuteron-induced inclusive breakup reaction in light and medium-mass nuclei', *Phys. Rev. C* **108**, 014617 (2023).
- 19 Antonio M. Moro, **Jin Lei** and Edward C. Simpson, 'Modelling inclusive breakup: application to incomplete fusion', *Journal of Physics: Conference Series* **2340**, 012034 (2022).

- 20 K. Wang, Y. Y. Yang, V. Guimarães, D. Y. Pang, F. F. Duan, Z. Y. Sun, **Jin Lei**, G. Yang, S. W. Xu, J. B. Ma, Q. Liu, Z. Bai, H. J. Ong, B. F. Lv, S. Guo, X. H. Wang, R. H. Li, M. Kumar Raju, Z. G. Hu and H. S. Xu, 'Elastic scattering investigation of radioactive ^{13}B and ^{13}O projectiles on a ^{208}Pb target at intermediate energies', *Phys. Rev. C* **105**, 054616 (2022).
- 21 F. F. Duan, Y. Y. Yang, **Jin Lei**, K. Wang, Z. Y. Sun, D. Y. Pang, J. S. Wang, X. Liu, S. W. Xu, J. B. Ma, P. Ma, Z. Bai, Q. Hu, Z. H. Gao, X. X. Xu, C. J. Lin, H. M. Jia, N. R. Ma, L. J. Sun, D. X. Wang, G. Yang, S. Y. Jin, Z. Z. Ren, Y. H. Zhang, X. H. Zhou, Z. G. Hu and H. S. Xu, 'Elastic scattering and breakup reactions of neutron-rich nucleus ^{11}Be on ^{208}Pb at 210 mev', *Phys. Rev. C* **105**, 034602 (2022).
- 22 O. C. B. Santos, R. Lichtenthäler, K. C. C. Pires, U. Umbelino, E. O. N. Zevallos, A. L. de Lara, A. S. Serra, V. Scardulli, J. Alcántara-Núñez, V. Guimarães, A. Lépine-Szily, J. C. Zamora, A. M. Moro, S. Appannababu, M. Assunção, A. Barioni, R. Linares, V. A. B. Zagatto, P. N. de Faria, M. C. Morais, V. Morcelle, J. M. B. Shorto and **Jin Lei**, 'Evidence of the effect of strong stripping channels on the dynamics of the $^8\text{Li} + ^{58}\text{Ni}$ reaction', *Phys. Rev. C* **103**, 064601 (2021).
- 23 K. Wang, Y. Y. Yang, A. M. Moro, V. Guimarães, **Jin Lei**, D. Y. Pang, F. F. Duan, J. L. Lou, J. C. Zamora, J. S. Wang, Z. Y. Sun, H. J. Ong, X. Liu, S. W. Xu, J. B. Ma, P. Ma, Z. Bai, Q. Hu, X. X. Xu, Z. H. Gao, G. Yang, S. Y. Jin, Y. H. Zhang, X. H. Zhou, Z. G. Hu and H. S. Xu, 'Elastic scattering and breakup reactions of the proton drip-line nucleus ^8B on ^{208}Pb at 238 mev', *Phys. Rev. C* **103**, 024606 (2021).
- 24 M. Gómez-Ramos, J. Gómez-Camacho, **Jin Lei** and A. M. Moro, 'The Hussein–McVoy formula for inclusive breakup revisited: A Tribute to Mahir Hussein', *Eur. Phys. J. A* **57**, 57 (2021).
- 25 **Jin Lei** and Angela Bonaccorso, 'Comparison of semiclassical transfer to continuum model with ichimura-austern-vincent model in medium energy knockout reactions', *Physics Letters B* **813**, 136032 (2021).
- 26 R. Spartà, A. Di Pietro, P. Figuera, O. Tengblad, A.M. Moro, I. Martel, J.P. Fernández-García, **Jin Lei**, L. Acosta, M.J.G. Borge, G. Bruni, J. Cederkäll, T. Davinson, J.D. Ovejás, L.M. Fraile, D. Galaviz, J. Halkjaer Jensen, B. Jonson, M. La Cognata, A. Perea, A.M. Sánchez-Benítez, N. Soić and S. Viñals, 'Probing proton halo effects in the $8\text{b}+64\text{zn}$ collision around the coulomb barrier', *Physics Letters B* **820**, 136477 (2021).
- 27 L. Yang, C.J. Lin, H. Yamaguchi, **Jin Lei**, P.W. Wen, M. Mazzocco, N.R. Ma, L.J. Sun, D.X. Wang, G.X. Zhang, K. Abe, S.M. Cha, K.Y. Chae, A. Diaz-Torres, J.L. Ferreira, S. Hayakawa, H.M. Jia, D. Kahl, A. Kim, M.S. Kwag, M. La Commara, R. Navarro Pérez, C. Parascandolo, D. Pierroutsakou, J. Rangel, Y. Sakaguchi, C. Signorini, E. Strano, X.X. Xu, F. Yang, Y.Y. Yang, G.L. Zhang, F.P. Zhong and J. Lubian, 'Insight into the reaction dynamics of proton drip-line nuclear system $17\text{f}+58\text{ni}$ at near-barrier energies', *Physics Letters B* **813**, 136045 (2021).
- 28 Calvin W Johnson, Kristina D Launey, Naftali Auerbach, Sonia Bacca, Bruce R Barrett, Carl R Brune, Mark A Caprio, Pierre Descouvemont, W H Dickhoff, Charlotte Elster, Patrick J Fasano, Kevin Fosse, Heiko Hergert, Morten Hjorth-Jensen, Linda Hlophe, Baishan Hu, Rodolfo M Id Betan, Andrea Idini, Sebastian König, Konstantinos Kravvaris, Dean Lee, **Jin Lei**, Alexis Mercenne, Rodrigo Navarro Perez, Witold Nazarewicz, Filomena M Nunes, Marek Płoszajczak, Jimmy Rotureau, Gautam Rupak, Andrey M Shirokov, Ian Thompson, James P Vary, Alexander Volya, Furong Xu, Remco G T. Zegers, Vladimir Zelevinsky and Xilin Zhang, 'White paper: from bound states to the continuum', *Journal of Physics G: Nuclear and Particle Physics* **47**, 123001 (2020).
- 29 **Jin Lei** and Pierre Descouvemont, 'Lagrange-mesh R -matrix method for inhomogeneous equations', *Phys. Rev. C* **102**, 014608 (2020).
- 30 F.F. Duan, Y.Y. Yang, K. Wang, A.M. Moro, V. Guimarães, D.Y. Pang, J.S. Wang, Z.Y. Sun, **Jin Lei**, A. Di Pietro, X. Liu, G. Yang, J.B. Ma, P. Ma, S.W. Xu, Z. Bai, X.X. Sun, Q. Hu, J.L. Lou, X.X. Xu, H.X. Li, S.Y. Jin, H.J. Ong, Q. Liu, J.S. Yao, H.K. Qi, C.J. Lin, H.M. Jia, N.R. Ma, L.J. Sun, D.X. Wang,

- Y.H. Zhang, X.H. Zhou, Z.G. Hu and H.S. Xu, ‘Scattering of the halo nucleus ^{11}Be from a lead target at 3.5 times the coulomb barrier energy’, *Physics Letters B* **811**, 135942 (2020).
- 31 J. P. Fernández-García, A. Di Pietro, P. Figuera, J. Gómez-Camacho, M. Lattuada, **Jin Lei**, A. M. Moro, M. Rodríguez-Gallardo and V. Scuderi, ‘Breakup mechanisms in the $^6\text{He} + ^{64}\text{Zn}$ reaction at near-barrier energies’, *Phys. Rev. C* **99**, 054605 (2019).
 - 32 L. Hlophe, **Jin Lei**, Ch. Elster, A. Nogga, F. M. Nunes, D. Jurčiukonis and A. Deluva, ‘Deuteron- α scattering: Separable versus nonseparable Faddeev approach’, *Phys. Rev. C* **100**, 034609 (2019).
 - 33 **Jin Lei** and Antonio M. Moro, ‘Puzzle of Complete Fusion Suppression in Weakly Bound Nuclei: A Trojan Horse Effect?’, *Phys. Rev. Lett.* **122**, 042503 (2019).
 - 34 **Jin Lei** and Antonio M. Moro, ‘Unraveling the Reaction Mechanisms Leading to Partial Fusion of Weakly Bound Nuclei’, *Phys. Rev. Lett.* **123**, 232501 (2019).
 - 35 Rodrigo Navarro Pérez and **Jin Lei**, ‘Is the unusual near-threshold potential behavior in elastic scattering of weakly-bound nuclei a precision error?’, *Physics Letters B* **795**, 200–205 (2019).
 - 36 A. Di Pietro, A.M. Moro, **Jin Lei** and R. de Diego, ‘Insights into the dynamics of breakup of the halo nucleus ^{11}Be on a ^{64}Zn target’, *Physics Letters B* **798**, 134954 (2019).
 - 37 **Jin Lei**, ‘Inclusive breakup calculations in angular momentum basis: Application to $^7\text{Li} + ^{58}\text{Ni}$ ’, *Phys. Rev. C* **97**, 034628 (2018).
 - 38 **Jin Lei**, L. Hlophe, Ch. Elster, A. Nogga, F. M. Nunes and D. R. Phillips, ‘Few-body universality in the deuteron- α system’, *Phys. Rev. C* **98**, 051001(R) (2018).
 - 39 **Jin Lei** and Antonio M. Moro, ‘Post-prior equivalence for transfer reactions with complex potentials’, *Phys. Rev. C* **97**, 011601(R) (2018).
 - 40 L. Hlophe, **Jin Lei**, Ch. Elster, A. Nogga and F. M. Nunes, ‘ ^6Li in a three-body model with realistic Forces: Separable versus nonseparable approach’, *Phys. Rev. C* **96**, 064003 (2017).
 - 41 **Jin Lei** and Antonio M. Moro, ‘Comprehensive analysis of large α yields observed in ^6Li -induced reactions’, *Phys. Rev. C* **95**, 044605 (2017).
 - 42 G. Potel, G. Perdikakis, B. V. Carlson, M. C. Atkinson, W. H. Dickhoff, J. E. Escher, M. S. Hussein, **Jin Lei**, W. Li, A. O. Macchiavelli, A. M. Moro, F. M. Nunes, S. D. Pain and J. Rotureau, ‘Toward a complete theory for predicting inclusive deuteron breakup away from stability’, *The European Physical Journal A* **53**, 178 (2017).
 - 43 Q. Ducasse, B. Jurado, M. Aïche, P. Marini, L. Mathieu, A. Görgen, M. Guttormsen, A. C. Larsen, T. Tornyi, J. N. Wilson, G. Barreau, G. Boutoux, S. Czajkowski, F. Giacoppo, F. Gunsing, T. W. Hagen, M. Lebois, **Jin Lei**, V. Méot, B. Morillon, A. M. Moro, T. Renstrøm, O. Roig, S. J. Rose, O. Sérot, S. Siem, I. Tsekhanovich, G. M. Tveten and M. Wiedeking, ‘Investigation of the $^{238}\text{U}(d, p)$ surrogate reaction via the simultaneous measurement of γ -decay and fission probabilities’, *Phys. Rev. C* **94**, 024614 (2016).
 - 44 **Jin Lei** and Antonio M. Moro, ‘Numerical assessment of post-prior equivalence for inclusive breakup reactions’, *Phys. Rev. C* **92**, 061602(R) (2015).
 - 45 **Jin Lei** and Antonio M. Moro, ‘Reexamining closed-form formulae for inclusive breakup: Application to deuteron- and ^6Li -induced reactions’, *Phys. Rev. C* **92**, 044616 (2015).
 - 46 Y. Y. Yang, J. S. Wang, Q. Wang, D. Y. Pang, J. B. Ma, M. R. Huang, P. Ma, S. L. Jin, J. L. Han, Z. Bai, **Jin Lei**, J. B. Chen, Q. Hu, R. Wada, S. Mukherjee, Z. Y. Sun, R. F. Chen, X. Y. Zhang, Z. G. Hu, X. H. Yuan, S. W. Xu, S. Z. Chen, X. G. Lei, L. X. Liu, W. H. Ma, S. T. Wang, D. Yan, X. H. Zhang, M. H. Zhao, Y. Zhou, Y. J. Zhou, Z. Y. Guo, Y. H. Zhang, H. S. Xu and G. Q. Xiao, ‘Quasi-elastic scattering of $^{10,11}\text{C}$ and ^{10}B from a ^{208}Pb target’, *Phys. Rev. C* **90**, 014606 (2014).

- 47 Y. Y. Yang, J. S. Wang, Q. Wang, D.Y Pang, J. B. Ma, M. R. Huang, J. L. Han, P. Ma, S. L. Jin, Z. Bai, Q. Hu, **Jin Lei**, J. B. Chen, N. Keeley, K. Rusek, R. Wada, S. Mukherjee, Z. Y. Sun, R. F. Chen, X. Y. Zhang, Z. G. Hu, X. H. Yuan, X. G. Cao, Z. G. Xu, S. W. Xu, C. Zhen, Z. Q. Chen, Z. Chen, S. Z. Chen, C. M. Du, L. M. Duan, F. Fu, B. X. Gou, J. Hu, J. J. He, X. G. Lei, S. L. Li, Y. Li, Q. Y. Lin, L. X. Liu, F. D. Shi, S. W. Tang, G. Xu, X. Xu, L. Y. Zhang, X. H. Zhang, W. Zhang, M. H. Zhao, Z. Y. Guo, Y. H. Zhang, H. S. Xu and G. Q. Xiao, 'Elastic scattering of the proton drip-line nucleus ^8B off a $^{\text{nat}}\text{Pb}$ target at 170.3 MeV', *Phys. Rev. C* **87**, 044613 (2013).
- 48 Y.Y. Yang, J.S. Wang, Q. Wang, J.B. Ma, M.R. Huang, J.L. Han, P. Ma, S.L. Jin, Z. Bai, Q. Hu, **Jin Lei**, J.B. Chen, R. Wada, Z.Y. Sun, R.F. Chen, X.Y. Zhang, Z.G. Hu, X.H. Yuan, X.G. Cao, Z.G. Xu, S.W. Xu, C. Zhen, Z.Q. Chen, Z. Chen, S.Z. Chen, C.M. Du, L.M. Duan, F. Fu, B.X. Gou, J. Hu, J.J. He, X.G. Lei, S.L. Li, Y. Li, Q.Y. Lin, L.X. Liu, F.D. Shi, S.W. Tang, G. Xu, X. Xu, L.Y. Zhang, X.H. Zhang, W. Zhang, M.H. Zhao, Y.H. Zhang and H.S. Xu, 'A method for the measurement of elastic scattering angular distribution at HIRFL-RIBLL', *Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment* **701**, 1–6 (2013).
- 49 **Jin Lei**, J. S. Wang, S. Mukherjee, Q. Wang and R. Wada, 'Phenomenological formula of total reaction cross sections for low-energy systems', *Phys. Rev. C* **86**, 057603 (2012).
- 50 **Jin Lei**, J. S. Wang, S. Mukherjee, Q. Wang, R. Wada, Y. Y. Yang, J. B. Chen, J. L. Han, M. R. Huang, Z. Bai, P. Ma, S. L. Jin, J. B. Ma, Y. Li and M. H. Zhao, 'Quarter-point angle for light, weakly bound projectiles', *Phys. Rev. C* **86**, 057602 (2012).

Conference Proceedings

- 1 A.M. Moro, J. Casal, **Jin Lei** and M. Gómez-Ramos, 'Reaction theory and advanced CDCC', *Journal of Physics: Conference Series* **1643**, 012100 (2020).
- 2 L. Hlophe, **Jin Lei**, Ch. Elster, A. Nogga and F. M. Nunes, 'Three-body approach to deuteron-alpha scattering using realistic forces in a separable or non-separable representation', *Recent Progress in Few-Body Physics*, edited by N. A. Orr, M. Ploszajczak, F. M. Marqués and J. Carbonell, 267–271 (2020).
- 3 F.M. Nunes, P.C. Capel, Ch. Elster, L. Hlophe, **Jin Lei**, Weichuan Li, A.E. Lovell, G. Potel, J. Rotureau and T. Poxon-Pearson, 'New developments in reaction theory: preparing for the frib era', *EPJ Web Conf.* **178**, 03001 (2018).
- 4 **Jin Lei** and Antonio M. Moro, 'Evaluation of inclusive breakup cross sections in reactions induced by weakly-bound nuclei within a three-body model', *EPJ Web of Conferences* **117**, 06016 (2016).
- 5 **Jin Lei** and Antonio M. Moro, 'Evaluation of Inclusive Breakup in Reactions Induced by Deuteron within a Three-Body Model', *Basic Concepts in Nuclear Physics: Theory, Experiments and Applications*, edited by José-Enrique García-Ramos, Clara E. Alonso, María Victoria Andrés and Francisco Pérez-Bernal, 207–208 (2016).
- 6 P. Marini, Q. Ducasse, B. Jurado, M. Aiche, L. Mathieu, G. Barreau, S. Czajkowski, I. Tsekhanovich, A. Moro, **Jin Lei**, F. Giacoppo, A. Gorgen, Torny, L. Audouin, L. Tassan-Got, J. N. Wilson, F. Gunsing, M. Guttormsen, A. C. Larsen, M. Lebois, T. Renstrom, S. Rose, S. Siem, G. M. Tveten, M. Wiedeking, O. Serot, G. Boutoux, V. Méot, B. Morillon, D. Denis-Petit, O. Roig, S. Oberstedt and A. Oberstedt, 'First simultaneous measurement of fission and gamma probabilities of ^{237}U and ^{239}Np via surrogate reactions', *EPJ Web of Conferences* **122**, 12004 (2016).
- 7 Marini, P., Ducasse, Q., Jurado, B., Aiche, M., Mathieu, L., Barreau, G., Czajkowski, S., Tsekhanovich, I., Moro, A., Lei, Jin, Giacoppo, F., Gorgen, A., Torny, Audouin, L., Tassan-Got, L., Wilson, J. N., Gunsing, F., Guttormsen, M., Larsen, A. C., Lebois, M., Renstrom, T., Rose, S., Siem, S., Tveten, G. M., Wiedeking, M., Serot, O., Boutoux, G., Méot, V., Morillon, B., Denis-Petit, D., Roig, O., Oberstedt, S. and Oberstedt, A., 'First simultaneous measurement of fission and gamma probabilities of ^{237}U and ^{239}Np via surrogate reactions', *EPJ Web of Conferences* **122**, 12004 (2016).

- 8 A. M. Moro, **Jin Lei**, M. Gómez-Ramos, J. M. Arias, R. de Diego, J. Gómez-Camacho and J. A. Lay, 'Recent Developments for the Calculation of Elastic and Non-elastic Breakup of Weakly-bound Nuclei', *Acta Phys. Polon.* **B47**, 821 (2016).
- 9 Antonio M. Moro and **Jin Lei**, 'Recent Advances in Nuclear Reaction Theories for Weakly Bound Nuclei: Reexamining the Problem of Inclusive Breakup', *Few-Body Systems* **57**, 319–330 (2016).
- 10 J. S. Wang, Y. Y. Yang, Q. Wang, **Jin Lei**, J. B. Ma, M. R. Huang, J. L. Han, P. Ma, S. L. Jin, Z. Bai, Q. Hu, J. B. Chen, R. Wada, Z. Y. Sun, R. F. Chen, X. Y. Zhang, Z. G. Hu, X. H. Yuan, X. G. Cao, Z. G. Xu, S. W. Xu, C. Zhen, Z. Q. Chen, Z. Chen, S. Z. Chen, C. M. Du, L. M. Duan, F. Fu, B. X. Gou, J. Hu, J. J. He, X. G. Lei, S. L. Li, Y. Li, Q. Y. Lin, L. X. Liu, F. D. Shi, S. W. Tang, G. Xu, L. Y. Zhang, X. H. Zhang, W. Zhang, M. H. Zhao, Y. H. Zhang, H. S. Xu, G. Q. Xiao, S. Mukhejee, N. Keeley, K. Rusek and D. Y. Pang, ' ^7Be , 8B^+ ^{208}Pb Elastic Scattering at Above-Barrier Energies', *Journal of Physics: Conference Series* **420**, 012075 (2013).

RESEARCH FUNDING

- 2026 – 2030 ◇ **Co-Investigator.** National Natural Science Foundation of China (Key Program), Grant No. 12535009, RMB 2,350,000 (PI: Prof. Z.Z. Ren).
- 2024 – 2028 ◇ **Principal Investigator.** National Natural Science Foundation of China (General Program), Grant No. 12475132, RMB 520,000.
- 2021 – 2023 ◇ **Principal Investigator.** National Natural Science Foundation of China (Youth Science Fund), Grant No. 12105204, RMB 300,000.

STUDENT ADVISING

Ph.D. Students

- Hao Liu ◇ Tongji University. First lead-author work: *Comparison of Ichimura-Austern-Vincent and Glauber models for the deuteron-induced inclusive breakup reaction*, *Phys. Rev. C* **108**, 014617 (2023).
- Junzhe Liu ◇ Tongji University. First lead-author work: *Testing the validity of the surface approximation for reactions induced by weakly bound nuclei*, *Phys. Rev. C* **108**, 024606 (2023).

M.Sc. Students

- Yazhou Lu ◇ Tongji University. First lead-author work: *Systematic single-folding optical potential for ^6Li and ^7Li based on $K\text{Doz}$ potentials*, *Phys. Rev. C* **108**, 024612 (2023).

TEACHING

- 2022-now ◇ Atomic Physics, Undergraduate course, Tongji University.
- 2019 ◇ Talent Course 6: Theory for exploring nuclear reaction experiments, June 3 to 21, 2019, Michigan State University, East Lansing, MI, USA

TEACHING (CONTINUED)

- ◇ Graduate course, Ohio University, 2019. On a few occasions, I helped Professor Charlotte Elster teach Physics 6021 : Quantum Mechanics.

MISCELLANEOUS EXPERIENCE

Editorial Service

2026 – Now ◇ Junior (Reviewing) Editor, *Chinese Physics C*.

Refereeing

- ◇ Referee for *Physical Review C*, *Chinese Physics C*, *European Physical Journal A*, *Nuclear Physics A*, and *Journal of Physics G*.

Professional Service

- 2020 ◇ Lead Organizer: Reaction Seminar, a special online seminar series for the COVID-19 period, more details can be found at <https://reactionseminar.github.io>
- 2021 ◇ Lead Organizer: Reaction Seminar 2021, a special online seminar series for the COVID-19 period, more details can be found at <https://reactionseminar2021.github.io>

REFERENCES

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